Event analysis: step-one checkpoint

NAUSICAA

The implemented checkpoint covers the versioned hit reader and Petit's deterministic binning, graph, backbone and truth-path functions. Physical arrays and arithmetic use Float32 in mm, MeV and ns. The reduced v3 files retain MC time and actual primary IDs; their values are verified against the original sources.

Validation

Julia suite	Passed assertions
Event core	1877
Event IO	92
Legacy ion analysis	61

The pinned Petit fixtures cover line, curved, branched, dense, lattice and separated geometries. Tie-free fixtures require matching graph edges and paths (up to reversal), with numerical tolerances for Float32 lengths and scores. Tied cases check valid neighbour choices, component sizes and energies, and path validity. The maximum accepted edge distance is explicit in every case.

Sample	Events	Hits	MB	Max relative error	Read ms
Double beta	8995	2833001	37.51	1.66e-06	1.12
Electron background	8483	2732400	35.94	1.86e-06	1.13

The energy error compares Float32 source-order event sums against independent Float64 sums over all events. The detector's reference relative energy resolution is 0.005. Read time is the mean of five indexed event reads after warming the reader, with the file already open; it depends on the machine and file cache.

Five events spread across each sample also exercise the deterministic stages on raw hits: 2 mm bins, kNN with k=10, 4 mm maximum distance and zero energy threshold. These are software checks. Scientific selection efficiencies require the subsequent ion-production and diffusion implementation. The approved production reconstruction prescription is b=3 sigma and d=6 sigma at every L.

Reproduction and next checkpoint

```
make validate-event-step1 PYTHON=/path/to/python3.11-or-newer
# Presentation only, from the existing HDF5:
make report-event-step1 PYTHON=/path/to/python3.11-or-newer
```

Numerical values and source hashes are saved in `results/event_analysis/step1_validation.h5`. The HTML review guide contains the implemented interfaces, docstrings and remaining design contracts. Implementation stops at this checkpoint.